

A MINIMUM-ORDER COUNTEREXAMPLE TO THE GENERALIZED ALTINIŞIK–ALTINTAŞ CUBE CONJECTURE

ABSTRACT. We exhibit a ten-element gcd-closed set whose LCM matrix is singular although its divisibility poset contains no order-embedded copy of the Boolean lattice B_3 . This order is best possible, and adjoining fresh primes gives such examples of every order $n \geq 10$. This settles the cases 10, 11, and 12 left open by Mattila, Haukkanen, and Mäntysalo for the unrestricted extension of the Altınışik–Altıntaş cube conjecture.

Let $S = \{x_1, \dots, x_n\}$ be a finite set of distinct positive integers. It is *gcd-closed* if $\gcd(x_i, x_j) \in S$ for all i, j , and its LCM matrix is

$$[S] = (\text{lcm}(x_i, x_j))_{i,j=1}^n.$$

A *cube subsemilattice* is a subsemilattice isomorphic to the Boolean lattice $B_3 = (2^{\{1,2,3\}}, \subseteq)$. We call a divisibility poset *cube-free* if there is no injection $\phi : B_3 \rightarrow S$ such that $X \subseteq Y$ if and only if $\phi(X) \mid \phi(Y)$. This implies, in particular, the absence of a cube subsemilattice.

Conjecture 4.2 of Altınışik and Altıntaş is stated only for nine-element gcd-closed sets [1, Conjecture 4.2]. Mattila, Haukkanen, and Mäntysalo proved that statement and then considered the unrestricted assertion

(GAA) $\det[S] = 0 \implies (S, \mid)$ contains a cube subsemilattice.

We call (GAA) the *generalized Altınışik–Altıntaş cube conjecture*. They produced a counterexample of order 13 and left the orders 10, 11, and 12 open [2, Sections 3–4].

Theorem 1. *For every integer $n \geq 10$, there is an n -element gcd-closed set S such that $[S]$ is singular and $(S, \mid)$ is cube-free. The minimum possible order is 10. One minimum-order example is*

$$S_0 = \{1, 2, 3, 5, 7, 14, 20, 21, 135, 3780\}.$$

Proof. Set

$$T = 3780, \quad A = \{2, 3, 5, 7\}, \quad M = \{14, 20, 21, 135\},$$

so that $S_0 = \{1\} \cup A \cup M \cup \{T\}$. Every element of S_0 divides T . The pairwise gcds between distinct elements of M are

$$\begin{aligned} \gcd(14, 20) &= 2, & \gcd(14, 21) &= 7, & \gcd(14, 135) &= 1, \\ \gcd(20, 21) &= 1, & \gcd(20, 135) &= 5, & \gcd(21, 135) &= 3. \end{aligned}$$

2020 *Mathematics Subject Classification.* Primary 11C20; Secondary 06A12.

Key words and phrases. LCM matrix, gcd-closed set, singular matrix, meet semilattice, Boolean lattice.

All remaining gcds are immediate, so S_0 is gcd-closed.

Define $\varepsilon_s = -1$ for $s \in \{1\} \cup M$ and $\varepsilon_s = 1$ for $s \in A \cup \{T\}$, and set

$$v_s = \frac{T\varepsilon_s}{s} \quad (s \in S_0).$$

In the displayed order of S_0 ,

$$v = (-3780, 1890, 1260, 756, 540, -270, -189, -180, -28, 1)^\top.$$

For every $x \in S_0$,

$$([S_0]v)_x = xT \sum_{s \in S_0} \frac{\varepsilon_s}{\gcd(x, s)}.$$

The sum vanishes for $x = 1$ because it is $-1 + 4 - 4 + 1$. If $x = a \in A$, then a divides exactly two elements of M , and the sum is

$$(-1 + 3 - 2) + \left(\frac{1}{a} - \frac{2}{a} + \frac{1}{a}\right) = 0.$$

If $x = m \in M$, let $a, b \in A$ be the two elements dividing m . Among the other three elements of M , one has gcd a with m , one has gcd b with m , and one is coprime to m . Hence the sum is

$$(-1 + 2 - 1) + \left(\frac{1}{a} - \frac{1}{a}\right) + \left(\frac{1}{b} - \frac{1}{b}\right) + \left(-\frac{1}{m} + \frac{1}{m}\right) = 0.$$

Finally, for $x = T$ the sum is

$$-1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} - \frac{1}{14} - \frac{1}{20} - \frac{1}{21} - \frac{1}{135} + \frac{1}{3780} = 0,$$

as is seen after multiplying by 3780. Thus $[S_0]v = 0$, so $[S_0]$ is singular.

The Hasse diagram of $(S_0, |)$ has levels $\{1\}$, A , M , and $\{T\}$. The cover graph between A and M is the eight-cycle

$$2 - 14 - 7 - 21 - 3 - 135 - 5 - 20 - 2.$$

Suppose that B_3 admitted an order embedding into $(S_0, |)$. Its bottom has seven strict upper elements and must therefore map to 1; dually, its top must map to T . Each atom of B_3 has three strict upper elements, so the three atoms must map into A . Each coatom has three strict lower elements, so the three coatoms must map into M . Thus one would obtain the atom-coatom incidence graph of B_3 by deleting one vertex from each part of the above eight-cycle. Those two deletions remove at least three edges, leaving at most five, whereas the atom-coatom incidence graph of B_3 is a six-cycle. This contradiction proves that $(S_0, |)$ is cube-free.

Let P be any finite set of distinct primes not dividing T , and put $S_P = S_0 \cup P$. Then S_P is gcd-closed. Extend v by zero on P . The rows indexed by S_0 remain zero, while for $p \in P$,

$$([S_P]v)_p = \sum_{s \in S_0} \text{lcm}(p, s)v_s = p \sum_{s \in S_0} sv_s = p([S_0]v)_1 = 0.$$

Hence $[S_P]$ is singular. Each $p \in P$ is maximal in $(S_P, |)$ and has only one strict lower element, namely 1. It cannot represent the top of an embedded

B_3 , which has seven strict lower elements, and every other element of B_3 has a strict upper element. Therefore S_P is cube-free. Taking $|P| = n - 10$ gives every order $n \geq 10$.

It remains to prove minimality. The nine-element theorem of Mattila, Haukkanen, and Mäntysalo excludes order 9. Suppose that a cube-free singular example S had $|S| < 9$. Let d be the gcd of all elements of S . Repeated gcd-closedness gives $d \in S$, and

$$S' = \{s/d : s \in S\}$$

is gcd-closed, contains 1, has the same divisibility poset, and satisfies $[S'] = d^{-1}[S]$. Choose a nonzero vector $w \in \ker[S']$ and adjoin $9 - |S|$ fresh primes, each coprime to every element of S' . Extending w by zero gives a kernel vector exactly as above, because the row indexed by 1 gives $\sum_{s \in S'} sw_s = 0$. The new primes cannot belong to an embedded cube, again by their upper and lower sets. This would produce a cube-free singular nine-element set, contradicting the cited theorem. Thus order 10 is minimum. $\square$

REFERENCES

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